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Nature-Inspired Feature Engineering and Cancer Detection Framework Leveraging Ensemble Convolutional Neural Architectures

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ABSTRACT: In general, diagnosing a brain tumor usually begins with magnetic resonance imaging (MRI). Once an MRI shows that there is a tumor in the brain, the most common way to determine the type of brain tumor is to look at the results from a sample of tissue after a biopsy or surgery. This study addresses the problems of segmentation and detection of abnormal brain tissues and normal tissues. Image processing is used in medical tools for the detection of tumors. Only MRI images are not able to identify the tumorous region. An improved semi-supervised clustering algorithm and morphological-based segmentation are proposed for the identification of tumor regions with the preprocessing of the image. Preprocessing is performed by a Median filter, and skull masking is used, followed by image enhancement using Contrast Limited Adaptive Histogram Equalization (CLAHE). The tumor segmentation process has been performed by the Semi-Supervised Fuzzy C-Means (SSFCM) clustering algorithm. After the tumor area is segmented, feature extraction is carried out by using the Gray Level Co-occurrence Matrix (GLCM). Then, to reduce the number of dimensions from features, the Artificial Bee Colony (ABC) algorithm is introduced, which increases the detection rate of the classifier. An Ensemble Convolution Neural Network (ECNN) detection system is used in an unsupervised manner, which will be used to create and maintain the pattern for future use. Also, for patterns, are have to find out the feature to train ECNN. The simulation results prove the significance in terms of quality parameters and accuracy in comparison to state-of-the-art techniques.

KEYWORDS: Brain Tumor, MRI Images, Cancer Detection, CLAHE, GLCM, ABC Optimization, ECCN.

1 Introduction

Brain tumors affect humans badly because of the abnormal growth of cells within the brain. It can disrupt proper brain function and be life-threatening. The brain controls all the necessary functions of the body. The most complex organ in the human body is the brain, and it is part of the central nervous system (CNS). The skull covers the brain, and the brain is composed of three matters: "gray matter (GM), white matter (WM), and cerebrospinal fluid (CSF)." CSF is a transparent liquid that encapsulates the brain and the spinal cord and provides many functions to the CNS. It provides a barrier against shocks, which consists of glucose, oxygen, and ions. It is distributed throughout the nervous tissue. CSF helps in the removal of waste products from nervous tissues [1]. According to the World Health Organization (WHO), there are 120 types of brain tumors. Broadly, brain tumors are categorized into two types: Primary brain tumors and Secondary brain tumors.

Primary Brain Tumors originate in the brain, and Secondary Brain Tumors do not originate in the brain but come into the brain through other body parts. This process of spreading the tumor is known as metastasis. Two types of brain tumors [2] have been identified as Malignant and Benign. Malignant tumors are fast-growing and cancerous. Benign tumors are slow-growing and non-cancerous, and less harmful than malignant tumors. In the medical field, brain tumors are characterized by their GRADE. The GRADE of a tumor can be decided according to its behavior under microscopic observation. GRADE I tumors are known as benign. Their look is nearly the same as normal brain cells, and their growth rate is slow compared to other grades.

GRADE I and GRADE II are known as low-grade tumors. GRADE III and GRADE IV are known as high-grade tumors. Their behavior is much different from normal cells and requires urgent treatment to cure. These types of tumors show the fastest growth rate and have the worst abnormal behavior than other grades. GRADE 1 and GRADE 2 are slow-growing, with fewer chances to come back. Do not spread in and need only surgery, other parts not Radiotherapy. GRADE 3 and GRADE 4 are fast-growing, come back even after surgery, spread to other parts, and Need Radiotherapy and Chemotherapy [3].

It is important to detect the brain tumor at an earlier stage to reduce the death rate, so most of the researchers suggested that the brain imaging techniques help radiologists and researchers to detect problems in the human brain, without the need for neurosurgery. There are a number of techniques available in hospitals throughout the world that have been proven to be safe [4-7]. (A) CT scan: Computed Tomography (CT) is used to construct brain images through a series of X-ray scans. (B) PET Scan: A Positron Emission Tomography (PET) scan is used to get a functional view of the brain. (C) MRI Scan In the last few years, the use of Magnetic Resonance Imaging (MRI) scanners in the medical field has grown superbly [8].

Doctors may use MRI scans in diagnosing brain tumors and cancer. An MRI scan is an efficient way to look inside the human body without the need to cut body. In an MRI system, the strength of the magnetic field is measured in terms of tesla and gauss (1tesla=10000gauss). Most of the medical applications use 0.5 to 2.0 tesla. Magnetic Resonance Imaging (MRI) is widely used as it provides greater contrast images of the brain and cancerous tissues compared with other medical imaging techniques. In MRI Modalities, the tissue appearance can be affected by the variable behavior of protons shown in different tissues. The speed at which mobile hydrogen protons are moving becomes helpful in determining the amount of signal produced by specific tissue [9]. This technique is basically used to detect the differences in the tissues, which is a far better technique as compared to computed tomography (CT). So, this makes this technique a special one for brain tumor detection. From the MRI images, the information such as the tumor's location provided the radiologists, an easy way to diagnose the tumor and plan the surgical approach for its removal.

Previously, various techniques or algorithms were used in image processing techniques to identify the brain tumor by using MRI images, but it is very important to get an accurate result, so this work will be used to improve the accuracy for identifying the brain tumor by using MRI images. Therefore, brain tumor identification can be done through MRI images by using image processing techniques to find tumors in early stages. This paper focuses on the identification of brain tumors using image processing techniques. The work consists of different stages. First stage of this work is preprocessing, consists of grey level conversion, noise gets removal using median filter and skull masking followed by image enhancement will be takes place. Secondly, brain tumor segmentation by a semi-supervised Fuzzy C-Means clustering algorithm. The third stage will be featuring extraction using the GLCM feature extraction method. The fourth stage is dimension reduction using the ABC algorithm, and the sixth stage is detection of a brain tumor using an Ensemble Convolution Neural Network classifier. These stages are explained briefly in the proposed methodology.

2 Literature Review

Badran et al [10] proposed a computer-based method for defining the tumor region in the brain using MRI images. The classification of the brain into a healthy brain or a brain having a tumor is first done, which is then followed by further classification into benign or malignant tumors. The algorithm incorporates steps for preprocessing, image segmentation, feature extraction, and image classification using neural network techniques. Finally, the tumor area is specified by the region of interest technique as a confirmation step.

Sonavane and Sonar [11] proposed a method for classification of Brain tumors using a Classifier and a Neural Network that uses segmentation algorithms, which include Skull stripping along with morphological operations, aiming at preprocessing. Anisotropic Diffusion Filtering is used for noise removal and smoothing, with edge detection locating the boundaries of the image. Secondly, wavelet transforms are used in the decomposition and the feature extraction of MRI, and in combining features of MRI images. Finally, Adaboost uses classification techniques, which learn by selecting discriminative features, proposed to classify MRI images into tumorous (malignant) and non-tumorous (benign).

Selvapandian and Manivannan [12] surveyed various conventional methodologies for brain tumor detection, and segmentation methodologies are presented with their tumor segmentation performance evaluation parameters. Conventional methods are classified as feature-based tumor classification systems and classifier-based tumor classification systems.

Sivakumar and Ganeshkumar [13] proposed a method that consists of preprocessing, feature extraction, and classification. The brain image is enhanced using the Heuristic histogram equalization technique. Then, texture and morphological features are extracted from the preprocessed image. These features are optimized using a Genetic Algorithm and trained and classified using an ANFIS classifier. The performance of the proposed ischemic stroke detection system is analyzed in terms of sensitivity, specificity, accuracy, positive predictive value, and negative predictive value.

Shree and Kumar [14] concentrated on a noise removal technique, the extraction of gray-level co-occurrence matrix (GLCM) features, and DWT-based brain tumor region growing segmentation to reduce the complexity and improve the performance. This was followed by morphological filtering, which removes the noise that can be formed after segmentation. The probabilistic neural network classifier was used to train and test the performance accuracy in the detection of tumor location in brain MRI images.

Sugapriya [15] proposed a new cooperative scheme that applies a semi-supervised fuzzy clustering algorithm. Specifically, the Otsu (Oral Tracheal Stylet Unit) method is used to remove the Background area from a Magnetic Resonance Image. Finally, the Semi-supervised Entropy Regularized Fuzzy Clustering algorithm (SER-FCM) is applied to improve the quality level. The intensity, shape deformation, symmetry, and texture features were extracted from each image. The usefulness and significance of this research are fully demonstrated within the extent of real-life application.

Bagri & Johari [16] compared the combination of texture and shape features using with texture Gray Level Co-occurrence Matrix and Hu-moments, and the combination of Tamura texture and shape invariant Hu-moments. For the performance evaluation of the system, we use the most commonly used methods, namely precision and recall.

Prasartvit et al [17] propose a novel method for dimension reduction based on artificial bee colony (ABC). The method employs swarm intelligence based on the bee foraging model in order to select features that allow us to generate subsets of dimensions from the original high-dimensional data, while the resulting subsets satisfy the defined objective. Support vector machine (SVM) is used in this study as a fitness evaluation of ABC in classification problems. To evaluate our method, we tested it with five datasets and compared it with other dimension reduction algorithms. The result of this study shows that using ABC and SVM is suitable for reducing the dimension of data.

Nanni et al [18] proposed a system that represents a very simple yet effective way of boosting the performance of trained CNNs by composing multiple CNNs into an ensemble and combining scores by the sum rule. Several types of ensembles are considered, with different CNN topologies along with different learning parameter sets. The proposed system not only exhibits strong discriminative power but also generalizes well over multiple datasets thanks to the combination of multiple descriptors based on different feature types, both learned and handcrafted. Separate classifiers are trained for each descriptor, and the entire set of classifiers is combined by the sum rule.

3 Proposed Methodology

In this work, brain tumor detection is done by an image processing technique. The main aim of this work is to detect the early stage of a brain tumor. Here, the input image is sent to preprocessing. Two processes are taking place in the preprocessing region.

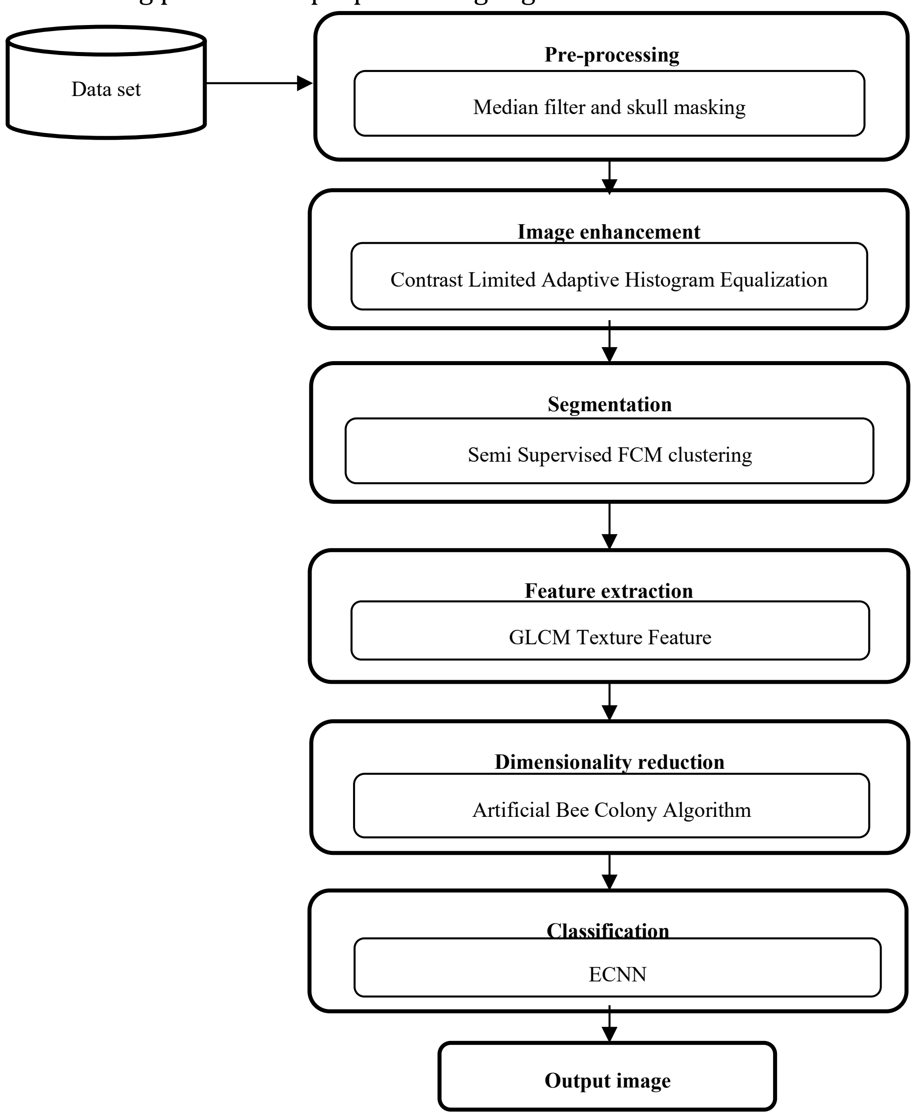

Figure 1: Overview of proposed methodology

The first one is noise removal using a median filter and skull masking, followed by image enhancement. To improve the performance and to reduce the complexity semi semi-supervised-based segmentation takes place. Feature extraction is done by using a texture feature extraction method, followed by feature dimension reduction to reduce the space by using an enhanced bee colony optimization algorithm. The brain tumor detection is done by using an Ensemble Convolution Neural Network classifier in an unsupervised manner. Figure 1. Shows a clear overview of the proposed methodology.

3.1 Pre-processing

In image acquisition, the image can be passed through whatever processes need to modify it or to extract information from it. The images are obtained from an MRI Scan of the brain. Different formats like JPG, PNG, etc. are used for storing the digital images obtained from the MRI of the brain. These images are stored in matrix form in MATLAB. MRI Scan images may be in RGB format. So RGB to grey conversion takes place here. Then the Median filter is used to denoising the MRI images by converting the image. Then, skull masking is the process that will be performed on the denoised image. The purpose of this process is to remove fatty tissue, skull part, and hair part in the MRI so we can process purely brain tissues only, and the ambiguity in the identification of tumors is reduced. For morphological filtering, we are using different masks on the MRI image: horizontal, diagonal, anti-diagonal, and vertical masks are used to process the skull masking. The functions performed by the preprocessing process are:

a. Noise removing – median filter.
b. Skull masking and
c. Image enhancement – CLAHE.

3.1.1 Median filter

This is the most common technique which used for noise elimination [19]. It is a 'non-linear' filtering technique. This is used to eliminate 'Salt and Pepper noise' from the greyscale image [20]. The median filter is based on the average value of pixels. The advantages of the median filter are efficient in reducing Salt and Pepper noise and Speckle noise. Also, the edges and boundaries are preserved. The median filter is a nonlinear signal processing technology based on statistics. The noisy value of the digital image or the sequence is replaced by the median value of the neighborhood (mask). The pixels of the mask are ranked in the order of their gray levels, and the median value of the group is stored to replace the noisy value. The median filtering output is $g(x, y) = med\{ f (x - i, y - j), i, j \in W\}$, where f (x, y), g(x, y) are the original image and the output image respectively, W is the two-dimensional mask: the mask size is n x n(where n is commonly odd) such as 3 x 3, 5 x 5, etc.; the mask shape may be linear, square, circular, cross, etc.

The noise-reducing performance of the median filter

The mathematical analysis is relatively complex for the image with random noise because the median filter is a nonlinear filter [21]. For an image with zero mean noise under a normal distribution, the noise variance of the median filtering is approximately,

$$\sigma_{med}^2 = \frac{1}{4nf^2(\bar{n})} \approx \frac{\sigma_i^2}{n+\frac{\pi}{2}-1}.\frac{\pi}{2} \quad (1)$$

where σ_i^2 is input noise power (the variance), n is the size of the median filtering mask, and f (n) is the function of the noise density. And the noise variance of the average filtering is,

$$\sigma_0^2 = \frac{1}{n}\sigma_i^2 \quad (2)$$

Comparing (1) and (2), the median filtering effects depend on two things: the size of the mask and the distribution of the noise. The median filtering performance of random noise reduction is better than the average filtering performance, but for the impulse noise, especially when narrow pulses are farther apart and the pulse width is less than n / 2, the median filter is very effective. It will be followed by a skull masking process using a morphological-based operation.

3.1.2 Skull masking

It means removal of non-brain tissues like skull, scalp, fat, eyes, neck, etc, from the MRI brain image. It helps to improve the speed and accuracy of the system in medical applications. Skull stripping is an important process in biomedical image analysis, and it is required for the effective examination of brain tumors from the MR images. By skull stripping, it is possible to remove additional cerebral tissues such as fat, skin, and skull in the brain images [22]. Skull stripping methods are broadly classified into five categories: mathematical morphology-based methods, intensity-based methods, deformable surface-based methods, atlas-based methods, and hybrid methods. In this work, skull stripping is done based on a mathematical morphological method.

Morphology-Based Methods

To analyze geometrical structures such as size, shape, connectivity, mathematical morphology is a tool used, which is based on set theory. It was developed originally for binary images but can be used for grayscale and color images. The procedure is to search the image at different locations with a pre-defined shape and to decide how this shape fits or misses the shapes in the image. This is called the structuring element, and it is also a binary image. Choice of size and shape of the structuring element is need-based. Diamond, square, disc, horizontal line, vertical line, cross, etc., are the most commonly used structuring elements. Erosion and dilation are the two essential operations in mathematical morphology.

Erosion and dilation

Let I be the binary image. S is the structuring element. Erosion of I by the structuring element S is denoted by IΘS. It denotes the set of all pixels for which S placed at that pixel is contained within I. Erosion shrinks or performs thinning of the object. It excludes all small unwanted objects in the image.

$$I\Theta S=\{z \mid (S)\, z \subseteq I\} \quad (3)$$

Dilation of I by the structuring element S is denoted by I ⊕ S. I overlaps by at least one element. Dilation performs thickening of an image. It highlights small objects in the image.

$$I \oplus S= \{ z \mid (S)\, z \cap I \neq \phi\} \quad (4)$$

After the process of morphological-based operation, the image enhancement process will take place using CLAHE for the improvement of the segmentation process.

3.1.3 Image enhancement- Contrast Limited Adaptive Histogram Equalization (CLAHE)

Adaptive histogram equalization is a computer image processing technique used to improve contrast in images. It differs from ordinary histogram equalization in the respect that the adaptive method computes several histograms, each corresponding to a distinct section of the image, and uses them to redistribute the lightness values of the image. Ordinary histogram equalization simply uses a single histogram for an entire image.

Consequently, adaptive histogram equalization is considered an image enhancement technique capable of improving an image's local contrast, bringing out more detail in the image. However, it can also produce significant noise. A generalization of adaptive histogram equalization called contrast-limited adaptive histogram equalization, also known as CLAHE, was developed to address the problem of noise amplification. CLAHE operates on small regions in the image, called tiles, rather than the entire image. Each tile's contrast is enhanced so that the histogram of the output region approximately matches the histogram specified by the 'Distribution' parameter. The neighbouring tiles are then combined using bilinear interpolation to eliminate artificially induced boundaries. The contrast, especially in homogeneous areas, can be limited to avoid amplifying any noise that might be present in the image. After image enhancement, segmentation will take place by using the SSFCM algorithm. Segmenting makes an image easier to further analyze and recognize important information from a digital image.

3.2 Segmentation

Image Segmentation is the procedure of partitioning a digital image into numerous regions or sets of pixels. Essentially, in image partitions are different objects have the same texture or color. The image segmentation results are a set of regions that cover the whole image together and a set of contours extracted from the image. All of the pixels in a region are similar with respect to some characteristics such as color, intensity, or texture. Neighboring regions are considerably different with respect to the same individuality. In our work, segmentation is done by using SSFCM will be discussed in the following section.

3.2.1 Semi-Supervised Fuzzy C-Means clustering algorithm (SSFCM)

There are different types of SSFCM [24] algorithms that differ in similarity and dissimilarity measurements, scaling factor, and optimization function. In this research, the proposed SSFCM algorithm is used to cluster the image. The main objective function of the SSFCM algorithm involves two supervised and unsupervised parts that are expressed in the following equation:

$$\mathrm{J} = \sum_{i=1}^{C} \sum_{j=1}^{N} u_{ij}^{p} d_{ij}^{2} + \alpha \sum_{i=1}^{c} \sum_{j=1}^{N} (u_{ij} - f_{ij} b_j)^p d_{ij}^{2} \tag{5}$$

Where u_{ij} is the membership value of data pattern j in cluster i, d_{ij} is the distance between the center of cluster v_i and data pattern j, c is the number of clusters, f_{ij} is the membership value of training data pattern j in cluster I, which never gets updated in the algorithm, b_j is a Boolean vector indicating that a particular pattern is training (3) or is not training (0), p is the fuzzifier parameter that is commonly set as 2 and α is a scaling factor that balances the supervised and unsupervised learning components. The scalding factor is usually expressed as a ratio (N/M) between the number of all data patterns (N) and training data patterns (M)[24].

The algorithm used in SSFCM is summarized in the following steps:

1. The first step is to determine the initial parameters, including the number of clusters c, initial membership values of training data patterns to each cluster (matrix F), vector b, and initial fuzzy partitioning matrix U^0 with random values between 0 and 1, where: {U $|_{uk} \in [0,1]$;

$$\sum_{i=1}^{C} u_{ik} = 1; 0 < \sum_{k=1}^{n} u_{ik} < n \tag{6}$$

2. Start the iterative procedure of the algorithm.

3. Calculate the cluster centers V=[v_i] (prototypes) as:

$$v_i = \frac{\sum_{j=1}^{N} u_{ij}^2 x_j}{\sum_{j=1}^{N} u_{ij}^2} \tag{7}$$

It should be noted that in calculating the initial cluster centers, only training data are used.

4. Update the partition matrix U:

$$u_{ij} = \frac{1}{1+\alpha}\left\{\frac{1+\alpha(1-b_j\sum_{l=1}^{C} f_{ij})}{\sum_{l=1}^{C}\left(\frac{d_{ij}}{d_{ij}}\right)^2} + \alpha f_{ij} b_j\right\} \tag{8}$$

5. And finally define the stopping criteria: If $J^r - J^{r-1} < \varepsilon$ stop the iteration; else repeat steps 2 to 5 by considering $U=U^r$. Equations for partitioning matrix (Eq. (8)) and prototypes (Eq. (7)) are derived by minimizing the objective function (Eq. (5)) using an optimization technique such as Lagrange multipliers. The minimization of the objective function is an iterative procedure that, for each iteration, calculates the membership matrix of non-training data, and the cluster's prototypes are calculated. The iterative procedure continues until the variation in the objective function in two consecutive iterations meets a termination criterion. Prototypes of each cluster and the membership of each sample to different clusters (partitioning matrix) are the outputs of the SSFCM clustering analysis.

Then, the feature extraction takes place by using GLCM for texture extraction, which extracts high-level features needed in order to perform classification of targets.

3.3 Feature extraction

The feature extraction [16] is extracting the cluster that shows the predicted tumor at the FCM output. Gray Level Co-occurrence Matrix (GLCM) is a statistical method for examining texture features that consider the spatial relationship of pixels. In this, a GLCM matrix is created by calculating how often a pixel with the intensity value i occurs in a specific spatial relationship to a pixel with the value j. GLCM consists of the frequencies at which two pixels are separated by a certain vector in the image. GLCM properties by which the distribution in the matrix will depend on the distance and angular or directions, like horizontal, vertical, diagonal, and anti-diagonal relationships between the pixels. Suppose an input image has M total number of pixels in horizontal directions and M total number of pixels in vertical directions. Suppose the gray level that appears at each pixel is quantized to z number of levels, assume N_x =1,2,3,…….M consists of horizontal space, and N_y=1,2,3…….N consists of vertical space, and G =0,1,2,3…Z consists of the set of Z quantized gray levels. In a given distance d and direction given by θ , the GLCM is calculated by using gray scale pixel i and j, expressed as the number of co-occurrence matrix in different directions.

$$P(i, j \mid d, \theta) = \frac{p(i,j \mid d,\theta)}{\sum_i \sum_j p(i,j \mid d,\theta)} \tag{9}$$

Among them are five features: Contrast, Correlation, Entropy, Energy, and Homogeneity.

3.3.1 Contrast

Contrast measures intensity between a pixel and its neighbor over the whole image, and it is considered zero for a constant image and it is also known as variance and moment of inertia.

$$Contrast = \sum_{i,j}(i-j)^2\, p(i,j) \tag{10}$$

3.3.2 Correlation

Correlation measures how a pixel is correlated to its neighbor over the whole image.

$$Correlation = \sum_{i,j}\frac{(1-\mu_i)(j-\mu_j)p(i,j)}{\delta_i \delta_j} \tag{11}$$

3.3.3 Entropy

Entropy gives a measure of the complexity of the image, and this complex texture tends to higher entropy.

$$Entropy = \sum_i \sum_j p(i,j) \tag{12}$$

3.3.4 Energy

Energy is the sum of squared elements in the GLCM, and it is by default one for a constant image.

$$Energy = \sum_{i,j} (i,j)^2 \tag{13}$$

Here, texture feature extraction is done by using GLCM Texture Feature for an efficient classification result. The next process is dimensionality reduction using the Artificial Bee Colony (ABC) algorithm.

3.3.5 Dimensionality Reduction

Dimensionality reduction is introduced to increase the detection rate of the classifier. It is done by using an Artificial Bee Colony (ABC). High dimensionality of data is a limiting factor to data processing in many fields. It causes ambiguity in identifying significant factors for data analysis. Dimension reduction is needed to separate irrelevant data from the desired data. This research proposes a novel method for dimension reduction based on ABC.

3.3.6 Artificial Bee Colony Optimization algorithm

Artificial Bee Colony (ABC) algorithm [17] simulates the foraging behaviour of bees. These specialized bees try to maximize the nectar amount stored in the hive by performing efficient division of self-organization. These bees are used to reduce the number of dimensions of the texture features. In order to perform this task, the ABC algorithm consists of three kinds of bees: Employed bees, onlooker bees, and scout bees. Half of the colony comprises employed bees, and the other half includes the onlooker bees. Employed bees are responsible for exploiting the information of texture features from the nectar sources explored before and giving feature information to the other waiting bees (onlooker bees) in the hive about the quality of the food source site, which they are exploiting. Onlooker bees wait in the hive and decide on a food source to exploit depending on the feature information shared by the employed bees. Scouts randomly search the environment in order to find a new food source, depending on features information. Let the bees select the feature subsets at random and calculate their fitness, and find the best one at each iteration. This procedure is repeated for several iterations to find the reduced feature subset.

In the first step of the algorithm, for each employed bee, whose total number is equal to half of the number of food sources, a new source is produced by:

$$v_{ij} = x_{ij} + \phi_{ij}(x_{ij} - x_{kj}) \tag{14}$$

where, φ_{ij} is a uniformly distributed real random number within the range [-1,1], k is the index of the solution chosen randomly from the colony (k = int (rand * N) + 1), j = 1,...D, and D are the dimensions of the problem. After producing v_i, this new reduced feature set is compared to solution x_i, and the employed bee exploits the better source. In the second step of the algorithm, an onlooker bee chooses a food source with a probability and produces a new reduced feature source at the selected food source site. As for the employed bee, the better source is decided based on the objective value (f_i). Fitness value has to be maximized, as given in the following equation:

$$\text{fiti} = \begin{cases} \frac{1}{1+f_i} & if\ f_i \geq 0 \\ 1 + abs(f_i) & otherwise \end{cases} \quad (15)$$

The probability is calculated by means of fitness value using the following equation:

$$\text{Pi} = \frac{fit_i}{\sum_{j=1}^{N} fit_j} \quad (16)$$

where, fit_i is the fitness of the solution x_i. After all onlookers are distributed to the sources, the sources are checked to determine whether they are to be abandoned. If the number of cycles that a source cannot be improved is greater than a predetermined limit, the source is considered to be exhausted. The employed bee associated with the exhausted source becomes a scout and makes a random search in the problem domain by the following equation:

$$\text{xij} = x_j^{min} + (x_j^{max} - x_j^{min}) * \text{r and} \quad (17)$$

Algorithm for Artificial Bee Colony for dimensionality reduction

C, the set of all conditional features;
D, the set of decision features.

(1) Select the initial parameter values for ABC.
(2) Initialize the population(x_i).
(3) Calculate the objective and fitness value.
(4) Find the optimum feature subset globally.
(5) Do.
a. Produce a new feature subset(v_i).
b. Apply the greedy selection between x_i and v_i.
c. Calculate the fitness and probability values.
d. Produce the solutions for onlookers.
e. Apply greedy selection for onlookers.
f. Determine the abandoned solution.
g. Calculate the cycle best feature subset.
h. Memorize the best optimum feature subset.
(6) Repeat // for the maximum number of cycles.

After the process of dimensionality reduction using the ABC algorithm, an ECNN will be considered as the last stage to identify the tumor and non-tumor images.

3.3.7 Algorithm Classification- Ensemble Convolution Neural Network (ECNN)

In this work, the classification is used to differentiate tumor and non-tumor regions by an Ensemble Convolution Neural Network classifier.

3.3.8 Ensemble Convolution Neural Network (ECNN) classifier

Ensemble is a process of combining the different classifier outputs into a single output to produce a better and efficient result. Here, various convolution neural networks will be combined using an ensemble method using the sum rule. To obtain a better estimator of the posterior probability by combining the resulting estimates of the individual members of the ensemble is the idea underlying multiple classifier fusion. CNNs are a class of deep feed-forward neural networks [18]. Like most neural networks, CNNs are composed of interconnected neurons that have inputs with learnable weights, biases, and activation functions. CNN layers have neurons arranged in three dimensions: width, height, and depth. This means that every layer in a CNN transforms a 3D input volume into a 3D output volume of neuron activations.

CNNs are built with five classes of layers: convolutional (CONV), activation (ACT), pooling (POOL), followed by a last stage, including Fully-Connected (FC), and classification (CLASS). The CONV layer is the core building block of a CNN and is also what makes CNNs so computationally expensive. These layers compute the outputs of neurons that are connected to local regions by applying a convolution operation to the input. The spatial extent of connectivity of these local regions is a hyperparameter called the receptive field, and a parameter sharing scheme is used in CONV Layers to control the number of parameters. This means that the parameters of CONV layers are shared sets of weights (also called kernels or filters) that have relatively small receptive fields.

POOL layers perform non-linear down-sampling operations. Max pooling is the most common non-linear operation: it partitions the input into a set of non-overlapping rectangles and outputs the maximum for each group. In this way, POOL reduces the spatial size of the representation while simultaneously reducing 1) the number of parameters, 2) the possibility of over-fitting, and 3) the computational complexity of the network. It is common practice to insert a POOL layer between CONV layers. ACT layers apply some activation functions, such as the non-saturating ReLU(Rectified Linear Unit) function $f(x) – \max(0,x)$ or the saturating hyperbolic tangent $f(x) – \tanh(x)$, $f(x) = |\tanh(x)|$, or the sigmoid function $f(x) = (1+e^{-x})^{-1}$.

FC layers have neurons that are fully connected to all the activations in the previous layer and are applied after CONV and POOL layers.

Finally, combine the results of all classifiers by using an average function. The average function is performed based on the arithmetic mean, giving the formula [27]

$$f_i^{am}(cl_i) = \frac{1}{N}\sum_{j=1}^{N} cl_i(j)\,, j = 1 \dots N \;\; no\ of\ samples, \tag{18}$$

where cl_i denotes the classification results from CNNs, i belongs to the number of classifiers, and j belongs to the number of samples; finally, classifier is used for the purpose to classify the tumor and non-tumor regions. Also, have the comparison on various classification methods, and the results are discussed in the following section briefly.

4 Results and Discussion

In this section, we are going to discuss various techniques with our proposed method. The dataset used here is the Brain Web dataset [28], which consists of full three-dimensional simulated brain MR data obtained using three sequences of modalities, namely, T1- T1-weighted MRI, T2-weighted MRI, and proton density-weighted MRI. This dataset included a variety of slice thicknesses, noise levels, and levels of intensity non-uniformity. The images used for our analysis are mostly included T2-weighted modality with 1 mm slice thickness, 3% noise, and 20% intensity non-uniformity.

In this dataset, 13 out of 44 images included are tumor-infected brain tissues. The last dataset collected from expert radiologists consisted of 135 images of 15 patients with all modalities. This dataset had ground truth images that helped to compare the results of the proposed method with the manual analysis of radiologists. Results of various classification techniques and the proposed method are measured in terms of sensitivity, specificity, and accuracy. The classification techniques used for the comparison are K-Nearest Neighbours(K-NN), Adaptive Fuzzy Inference System (ANFIS), Support Vector Machine (SVM), and Ensemble Convolution Neural Network (ECNN) [29]. The three types of aspects, namely sensitivity, specificity, and accuracy, are discussed briefly with a comparison between various classification methods in the following section.

4.1 Sensitivity

Sensitivity (also called the true positive rate, the recall, or probability of detection in some fields) measures the proportion of actual positives that are correctly identified as such (e.g., the percentage of sick people who are correctly identified as having the condition). It can be written as,

$$Sensitivity = \frac{True\ Positive}{True\ Positive + False\ Negative} \tag{19}$$

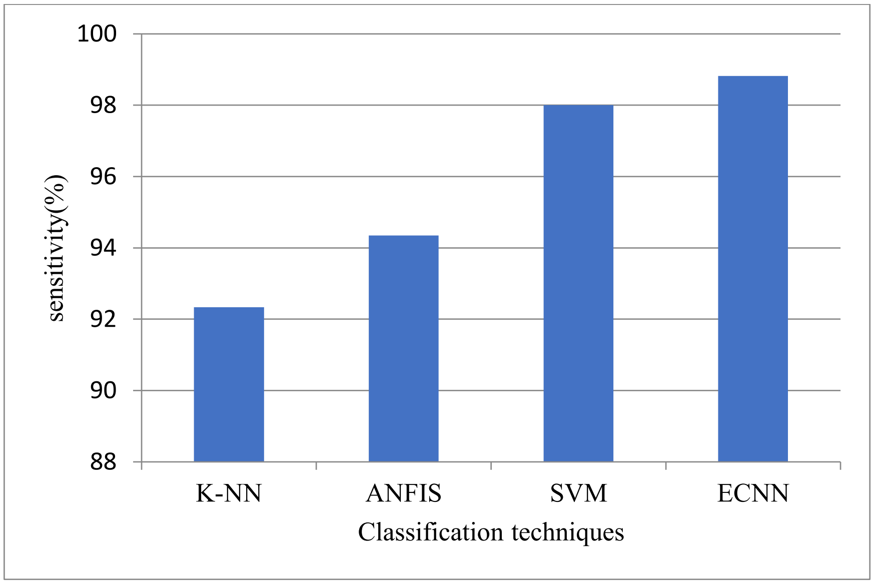

Figure 2: Comparison of various techniques with Sensitivity

Figure 2 shows the comparison of various classification techniques with sensitivity. It is clear that the proposed ECNN has 98.82% as sensitivity, and the existing K-NN, ANFIS, and SVM classification have 92.34%, 94.35% and 98% respectively.

4.2 Specificity

Specificity (also called the true negative rate) measures the proportion of actual negatives that are correctly identified as such (e.g., the percentage of healthy people who are correctly identified as not having the condition). It can be written as,

$$Specificity = \frac{True\ Negative}{True\ Negative + False\ Positive} \tag{20}$$

Figure 3 shows the comparison of various classification techniques with specificity. It is clear that the proposed ECNN has 88.89% a specificity and the existing K-NN, ANFIS, and SVM classification has a specificity value of 61.54%, 69.23% and 87.12% respectively.

4.3 Accuracy

The accuracy is the proportion of the total number of correct predictions. It is used to calculate the rate of the classifier by using the equation:

$$Accuracy = \frac{True\ Positive + True\ Negative}{Positive + Negative} \tag{21}$$

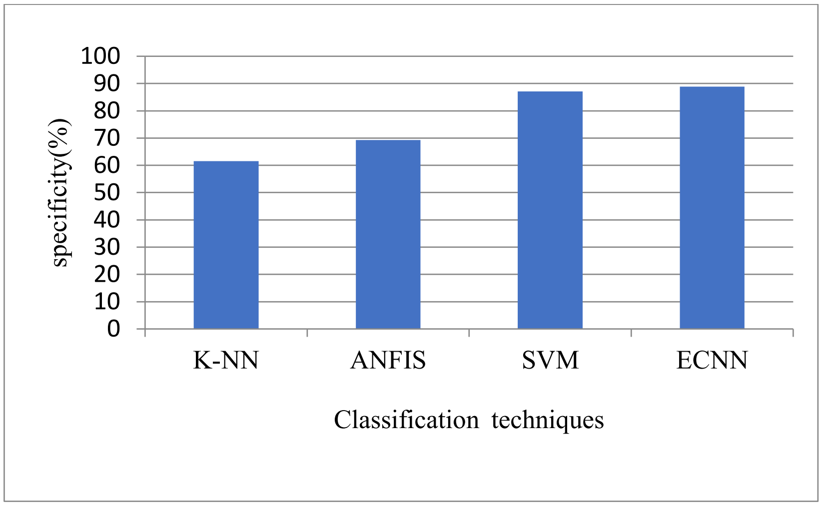

Figure 3: Comparison of various techniques with Specificity

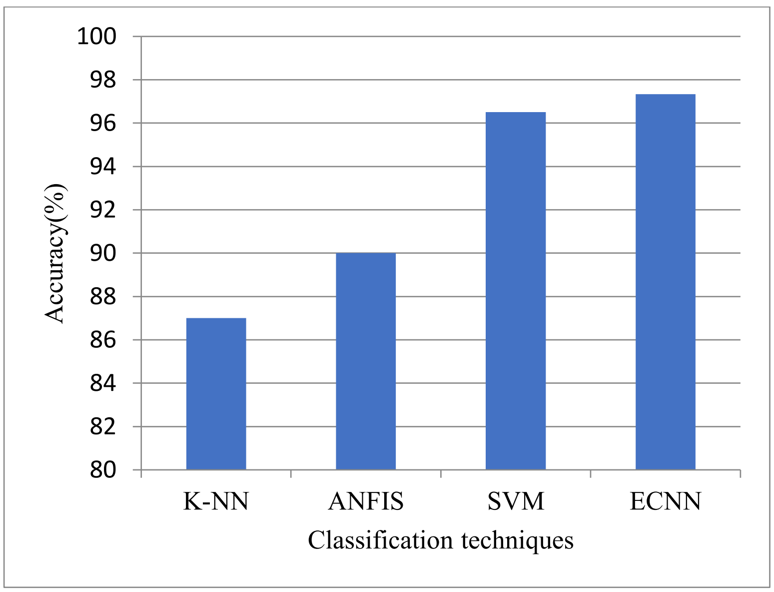

Figure 4: Comparison of various techniques with Accuracy

Figure 4 shows the comparison of various classification techniques with accuracy. It is clear that the proposed ECNN has 97.33% an accuracy and the existing K-NN, ANFIS, and SVM classification has accuracy values of 87%, 90% and 96.51% respectively. Hence, the proposed ECNN classifier has high accuracy, specificity, and sensitivity values when it is compared to all other existing work.

5 Conclusion

In this work, a tumor is detected by using a data mining process. Various stages are carried out for the detection of tumors. The stages included in this process are pre-processing, segmentation, feature extraction, and classification. Pre-processing consists of filtering, skull masking, and image enhancement by CLAHE for an efficient segmentation process. The important role of this work is feature texture extraction using GLCM and classification using Ensemble Convolution Neural Network (ECNN) for the detection of brain tumors, and it has a comparison between the proposed ECNN classifier with other existing methods, namely K-NN, ANFIS, and SVM, and proves that the proposed ECNN method has a high accuracy, sensitivity, and specificity value when it is compared with other existing classifiers.

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References

1. Toure ML, Beiji Z, Musau F, Camara AD. Advanced algorithm for brain segmentation using fuzzy to localize cancer and epilespy region. In 2010 International Conference on Electronics and Information Engineering; 2010 Aug 1, Kyoto, Japan;2010. Vol. 2, p.V2-488.
2. Kapse RS, Salankar SS, Babar M. Literature survey on detection of brain tumor from MRI images. IOSR-JECE. 2015;10(1):80-6.
3. Joshi MA, Shah DH. Survey of brain tumor detection techniques through MRI images. AIJRFANS ISSN. 2015:2328-3785.
4. Demirhan A, Törü M, Güler I. Segmentation of tumor and edema along with healthy tissues of brain using wavelets and neural networks.IEEE J Biomed Health Inform. 2014;19(4):1451-8.
5. Madhukumar S, Santhiyakumari N. Evaluation of k-Means and fuzzy C-means segmentation on MR images of brain. Egypt J Radiol Nucl Med. 2015;46(2):475-9.
6. Kong Y, Deng Y, Dai Q. Discriminative clustering and feature selection for brain MRI segmentation. IEEE Signal Process Lett. 2014;22(5):573-7.
7. El-Melegy MT, Mokhtar HM. Tumor segmentation in brain MRI using a fuzzy approach with class center priors. Eurasip J Image Video Process. 2014;2014(1):21.
8. Anitha V, Murugavalli SJ. Brain tumour classification using two‐tier classifier with adaptive segmentation technique. IET Comput Vis. 2016;10(1):9-17.
9. Poonam JP. Review of image processing techniques for automatic detection of tumor in human brain. Int J Mob Comput Multimed Commun. 2013;2(11):117-22.
10. Badran EF, Mahmoud EG, Hamdy N. An algorithm for detecting brain tumors in MRI images. In The 2010 International Conference on Computer Engineering & Systems; 2010 Nov 30, Cairo, Egypt; 2010. p. 368-373.
11. Sonavane R, Sonar P. Classification and segmentation of brain tumor using Adaboost classifier. In 2016 International Conference on Global Trends in Signal Processing, Information Computing and Communication (ICGTSPICC); 2016 Dec 22, Jalgaon, India; 2016. p. 396-403.
12. Selvapandian A, Manivannan K. A detailed survey on brain tumor detection and classification systems based on features and classifiers. J Appl Nat Sci. 2017;11(7):341-5.
13. Sivakumar P, Ganeshkumar P. An efficient automated methodology for detecting and segmenting the ischemic stroke in brain MRI images. Int J Imaging Syst Technol. 2017;27(3):265-72.
14. Varuna Shree N, Kumar TN. Identification and classification of brain tumor MRI images with feature extraction using DWT and probabilistic neural network. Brain Inform. 2018;5(1):23-30.
15. Sugapriya C. Semi supervised approach-based brain tumor detection with noise removal. Int J Recent Innov Trends Comput Commun. 2017;5(5):446-52.
16. Bagri N, Johari PK. A comparative study on feature extraction using texture and shape for content-based image retrieval. Int J Adv Sci Technol. 2015;80(4):41-52.
17. Prasartvit T, Kaewkamnerdpong B, Achalakul T. Dimensional reduction based on artificial bee colony for classification problems. In International Conference on Intelligent Computing; 2011 Aug 11, Berlin, Heidelberg; 2011. p. 168-175.

18. Nanni L, Ghidoni S, Brahnam S. Ensemble of convolutional neural networks for bioimage classification. Appl Comput Inform. 2021;17(1):19-35.
19. Telrandhe SR, Pimpalkar A, Kendhe A. Implementation of brain tumor detection using segmentation algorithm & SVM. Comput Sci Eng. 2016;8(7):278-84.
20. Rajesh, C.P. and Bhalchandra, A.S., 2012. Brain tumour extraction from MRI images using MATLAB. Int. J. Electron. Commun. Soft Comput. Sci. Eng, 2(1), pp.1-4.
21. Zhu Y, Huang C. An improved median filtering algorithm for image noise reduction. Phys Procedia. 2012;25:609-16.
22. Pydi V, Josyula SB. Detection of Brain Tumor Using PCA With SVM. Int J Inf Commun Technol Educ. 2015;4(8):535-39.
23. Vij K, Singh Y. Enhancement of images using histogram processing techniques. Int J Comp Tech Appl. 2009;2(2):309-13.
24. Fatehi M, Asadi HH. Application of semi-supervised fuzzy c-means method in clustering multivariate geochemical data, a case study from the Dalli Cu-Au porphyry deposit in central Iran. Ore Geol Rev. 2017;81:245-55.
25. Sharma R, Sharma R. Image segmentation using morphological operation for automatic region growing. IJIRCCE. 2014;2(9):5686-93.
26. Chai HY, Wee LK, Swee TT, Salleh SH, Ariff AK. Gray-level co-occurrence matrix bone fracture detection. Am J Appl Sci. 2011;8(1):26.
27. Alexandre LA, Campilho AC, Kamel M. On combining classifiers using sum and product rules. Pattern Recognit Lett. 2001;22(12):1283-9.
28. Brainweb: Simulated BrainDatabase," http://brainweb.bic.mni.mcgill.ca/cgi/brainweb1.
29. Bahadure NB, Ray AK, Thethi HP. Image analysis for MRI based brain tumor detection and feature extraction using biologically inspired BWT and SVM. Int J Biomed Imaging. 2017;2017(1):9749108.